

Emulsifiers are usually long chain compounds with polar groups for example-soap. The soap molecules consist of two parts- (i) long hydrocarbon chain ( $C_{17}H_{35}$ ) which is soluble in fat and (ii) dissolved polar parts in water ( $COO^- Na^+$ ). During emulsion, the dipolar alkaline group of soap dissolves oil or grease and the polar group get dissolved in water. When rubbed, the greases are suspended in the water in the form of small droplets and flows smoothly with water.

**3. What is detergent?**

- (a) Soap (b) Medicine  
(c) Catalyst (d) Refining agent

**40<sup>th</sup> B.P.S.C. (Pre) 1995**

**Ans. (d)**

A detergent is different from soap. A detergent is a surfactant or a mixture of surfactants with cleaning and refining properties in dilute solutions. The most important ingredients in detergents are chemicals called surfactants. The term detergent by itself refers specifically to laundry detergent or dish detergent as opposed to hand soap or other types of cleaning agents. Detergents are commonly available as a powder or concentrated solutions.

**4. Detergents used for cleaning clothes are :**

- (a) Carbonates  
(b) Bicarbonates  
(c) Bismuthates  
(d) Sulfonates  
(e) None of the above/More than one of the above

**63<sup>rd</sup> B.P.S.C. (Pre) 2017**

**Ans. (d)**

Detergents used for cleaning clothes are usually alkylbenzene sulfonates, a family of compounds that are similar to soap but are more soluble in hard water.

## The Fertilizers

**1. What is manganese used for?**

- (a) Matchbox industry  
(b) Photography industry  
(c) Battery production  
(d) More than one of the above  
(e) None of the above

**BPSC (Tre-3) {Class 6-8}-2024**

**Ans. (d)**

- Manganese is used in various industries, including the production of steel, batteries, and as a catalyst in the chemical industry.
- Manganese plays an important role in battery-powered products. It acts as a stabilizer in the structure of NMC (nickel manganese cobalt) cathode materials and cathode materials are the key performing drivers of Li-ion batteries.
- Manganese compounds are used in photography for their colour effects, but this is not the primary use. It is also used in matchbox industry but not the primary one.

**2. For wheat cultivation which fertilizer is used?**

- (a) Nitrogen (b) Potassium  
(c) Copper (d) Iron

**47<sup>th</sup> B.P.S.C. (Pre) 2005**

**Ans. (a)**

Nitrogen is an essential element of plants. For wheat cultivation, the nitrogen fertilizer is most important for its growth and development. The deficiency of nitrogen is probably the most common nutritional problem affecting the plants.

**3. The elements present in urea are :**

- (a) C, H, O (b) C, N, O  
(c) C, H, N (d) C, H, N, O  
(e) None of the above / More than one of the above

**66<sup>th</sup> B.P.S.C. (Pre) 2020**

**Ans. (d)**

Urea or Carbamide is an organic compound with a chemical formula  $CO(NH_2)_2$  having 46-47% nitrogen approximately. The urea molecule has two amide ( $NH_2$ ) groups containing nitrogen.

**4. Urea can be identified by**

- (a) Biurate test (b) Benedict test  
(c) Lucas test (d) Mulliken test  
(e) None of the above / More than one of the above

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**Ans. (a)**

A biochemical test to detect proteins in solution, named after the substance biuret ( $H_2NCONHCONH_2$ ), which is formed when urea is heated. Sodium hydroxide is mixed with the test solution and drops of 1% copper (II) sulphate solution are then added slowly. A positive result is indicated by a violet ring, caused by the reaction of peptide bonds in the proteins or peptides. Such a result will not occur in the presence of free amino acids.